

Sensations

Every sense does the same job. It turns some kind of energy into a nerve signal.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Explain what a receptor does, and name the five kinds
- Say why you stop noticing a smell but never stop noticing pain
- Trace light to the retina, and explain how the eye focuses up close
- Trace sound to the hair cells, and tell the two kinds of hearing loss apart
- Explain referred pain, and why it shows up in the wrong place

1 One job, five ways of doing it

A receptor is a translator. Whatever energy arrives, what leaves is always the same kind of signal.

Light, sound, pressure, chemicals, heat. All of it gets turned into a change in voltage. If that change is big enough, an action potential travels to the brain.

This conversion is called **transduction**, and every sense uses it.

RECEPTOR TYPE	RESPONDS TO	EXAMPLE
Mechanoreceptor	Being pushed or bent	Touch, pressure, hearing, balance, blood pressure
Chemoreceptor	Chemicals	Taste, smell, blood oxygen and carbon dioxide
Photoreceptor	Light	Rods and cones
Thermoreceptor	Temperature	Warm and cold receptors in skin
Nociceptor	Tissue damage	Pain, anywhere

The brain decides what a signal means, not the receptor. Action potentials look the same wherever they come from.

What makes light look like light is which part of the brain the signal reaches. Press on a closed eye and you see a flash. That is pressure, read as light, because it arrived on the visual pathway.

2 Adaptation, and the one sense that refuses

Most receptors stop reporting something that has not changed.

You feel your watch when you put it on, then stop feeling it. A smell fades within minutes of walking into a room.

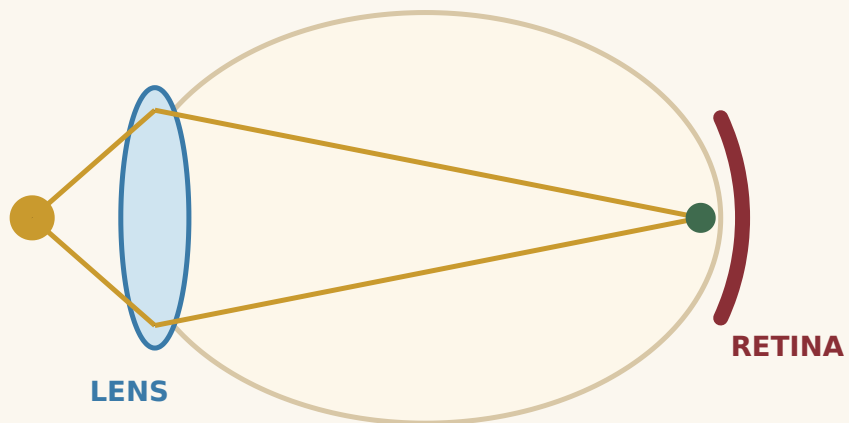
This is **sensory adaptation**, and it is useful. Information that never changes is not worth the bandwidth.

Pain receptors barely adapt at all. Pain that stopped being reported is pain you would stop reacting to, and reacting is the whole point. That is why a burn keeps hurting long after you moved away from the stove.

3 Vision — getting light onto the retina

Move the object closer and watch the lens work.

ACCOMMODATION — AND WHAT HAPPENS WHEN THE LENS STIFFENS



Distant object. The lens is relaxed and thin, and light lands on the retina without effort. This is the eye's resting state.

Light passes the cornea, then the pupil, then the lens, and lands on the **retina**.

The cornea does most of the bending, but only the lens can *change*. A muscle contracts, the lens gets rounder, and near objects come into focus. That is **accommodation**.

On the retina, **rods** work in dim light and give no color. **Cones** need bright light and give color and fine detail, and they are packed at the center.

That is why you see poorly in color at night, and why looking slightly to one side of a dim star makes it easier to see.

WHY THE NURSE CARES

From the forties on, the lens stiffens and stops rounding up. Near vision blurs first, and reading glasses follow. The eye is not failing. One tissue is just losing its stretch, which is what the button above copies.

4 Hearing — three steps and a spiral

Sound changes form twice on the way in. Where it fails tells you the type of deafness.

- 1 **Air.** Sound waves travel down the canal and shake the eardrum.
- 2 **Bone.** Three tiny bones amplify that shaking and push it into the fluid of the inner ear.
- 3 **Fluid.** Waves travel through the cochlea and bend hair cells, which fire the hearing nerve.

Different parts of the spiral respond to different pitches. So **which** hair cells bend is what makes a sound high or low.

Hair cells do not grow back. That is why damage from loud noise is permanent.

TYPE OF LOSS	WHERE THE PROBLEM IS	COMMON CAUSE
Conductive	Outer or middle ear. Sound cannot get in	Wax, fluid behind the eardrum, a bone problem
Sensorineural	Cochlea or nerve. Sound arrives but is not converted	Loud noise, aging, certain drugs

The inner ear also holds the balance organs. Fluid moving in the loops reports turning, and other parts report head position. That is why an inner ear infection can make the room spin.

5 Pain, and why it lies about location

Nerves from organs and nerves from skin arrive at the same parts of the spinal cord.

The brain has spent a lifetime learning that traffic on a given segment means skin. When the heart sends pain up that same segment, the brain reads it the usual way, and reports pain in the left arm and jaw.

That is **referred pain**. The map is worth knowing. The diaphragm refers to the shoulder tip. The gallbladder refers to the right shoulder blade. The heart refers to the left arm and jaw.

Taste and smell are wired together. Both use chemoreceptors. Most of what people call the taste of food is actually smell, arriving from the back of the mouth. That is why food seems flat with a blocked nose, even though the tongue works fine.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. A nurse stops noticing the smell of the unit after twenty minutes. What is this?

- a) Transduction
- b) Sensory adaptation
- c) Referred sensation
- d) Loss of chemoreceptors

2. Why does pressing on a closed eye produce a flash of light?

- a) Pressure makes light inside the eye
- b) The signal reaches the visual part of the brain, which reads anything there as light
- c) Photoreceptors respond to pressure as well as light
- d) The optic nerve glows when squeezed

3. A 50-year-old has to hold a menu at arm's length to read it. What changed?

- a) The retina lost its cones
- b) The lens stiffened and cannot round up for near vision
- c) The pupil no longer narrows
- d) The cornea flattened suddenly

4. A patient has hearing loss from fluid trapped behind the eardrum. What kind is it?

- a) Sensorineural loss
- b) Conductive loss
- c) Loss of the balance organs
- d) Damage to the hearing part of the brain

5. A patient having a heart attack feels pain in the left arm and jaw. Why?

- a) The heart pulls on the arm nerves
- b) Nerves from the heart and from the skin arrive at the same spinal segments
- c) Clots travel to the arm
- d) The heart's pain receptors are located in the arm

Answer key

1. **Sensory adaptation**

Correct. Receptors report less when a stimulus stops changing.

2. **The signal reaches the visual part of the brain, which reads anything there as light**

Right. Meaning comes from where the signal lands, not from the stimulus.

3. **The lens stiffened and cannot round up for near vision**

Correct. Focusing up close needs a lens that can change shape, and that ability fades with age.

4. **Conductive loss**

Right. Sound cannot get in, even though conversion would work fine if it arrived.

5. **Nerves from the heart and from the skin arrive at the same spinal segments**

Correct. The brain reads familiar traffic the familiar way and puts the pain in the wrong place.

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